

K.S.R.M. COLLEGE OF ENGINEERING (AUTONOMOUS), KADAPA
B.Tech IV Semester (R23UG) Reg. & Supple. Exams Jun. – Jul. 2025
OPERATING SYSTEMS
(CSE)

Time: 3 Hours Max. Marks: 70

- Instructions:
- 1. Answer all 10 questions from Part-A. Each question carries two marks
 - 2. Answer one full question from each unit in Part-B. Each full question carries 10 marks

PART-A			10*2 = 20M														
1	a	What are the key services provided by an operating system?	2M	CO1	L2												
	b	Differentiate between user mode and kernel mode in an OS.	2M	CO1	L3												
	c	What are the main criteria for CPU scheduling?	2M	CO2	L3												
	d	Explain how multithreading improves system performance.	2M	CO2	L5												
	e	Define mutex locks	2M	CO3	L2												
	f	How does a system recover from deadlock?	2M	CO3	L5												
	g	Define paging.	2M	CO4	L2												
	h	How does the Translation Lookaside Buffer (TLB) improve performance?	2M	CO4	L5												
	i	Differentiate between protection and security	2M	CO5	L3												
	j	List the different file operations	2M	CO5	L2												
PART-B			5*10=50M														
UNIT-I																	
2		Explain the importance of system calls and discuss their types with examples	10M	CO1	L2												
OR																	
3		Define operating system. Explain the various functions of operating systems.	10M	CO1	L2												
UNIT-II																	
4		Explain FCFS and Round Robin scheduling algorithms. Find the average waiting time and average turn around time for a process, if the following processes are scheduling using FCFS and round robin scheduling algorithms. Time quantum is 1 msec. <table><tr><td><u>Process</u></td><td><u>burst time</u></td></tr><tr><td>P1</td><td>10</td></tr><tr><td>P2</td><td>1</td></tr><tr><td>P3</td><td>2</td></tr><tr><td>P4</td><td>1</td></tr><tr><td>P5</td><td>5</td></tr></table>	<u>Process</u>	<u>burst time</u>	P1	10	P2	1	P3	2	P4	1	P5	5	10M	CO2	L3
<u>Process</u>	<u>burst time</u>																
P1	10																
P2	1																
P3	2																
P4	1																
P5	5																
(OR)																	

5	a	Analyze how multi-core processors affect CPU scheduling strategies	5M	CO2	L4
	b	Contrast shared memory and message passing	5M	CO2	L4
UNIT-III					
6		Explain the deadlock avoidance with the help of Banker's algorithm.	10M	CO3	L4
(OR)					
7	a	What is critical section? Write Peterson's solution for critical section problem.	5M	CO3	L3
	b	Choose monitors to find the solution for Dining Philosophers Problem	5M	CO3	
UNIT-IV					
8		Consider the following page-reference string:0,1,2,3,0,1,2,3,0,1,2,3,4,5,6,7. How many page faults occur for the following page replacement algorithms, assuming Frame count=3 i)FIFO ii) LRU iii) OPTIMAL	10M	CO4	L3
(OR)					
9	a	Explain the working of demand paging and page fault handling.	5M	CO4	L2
	b	What is contiguous memory allocation? Explain its types.	5M	CO4	L2
UNIT-V					
10	a	Explain Security Problems in detail	5M	CO5	L2
	b	Evaluate the security threats posed by different file system structures.	5M	CO5	L5
(OR)					
11	a	Compare different directory structures in a file system.	5M	CO5	L2
	b	Explain about access matrix.	5M	CO5	L2
